

Affective transfer between LLM agents: the estimator, two pre-registered ablations, and a control that bites

Manan Wadhwa*

Abstract

When one language-model agent is steered into an emotion and talks to another, the second agent’s internal representation of its own state moves with the dose: for five of six emotions on Qwen3.6-27B and the same five on Llama-3-8B. We ask what carries it, and find that the standard linear tools give wrong answers for reasons their usual checks do not catch. The logistic-regression direction most work uses to read and write such states reproduces across independent fits at only 0.57 (27B) and 0.29 (8B) while difference-of-means reaches 0.97 and 0.94 on the same splits, and holding the per-sample penalty fixed does not close the gap. The reason is that every logistic fit on these activations is a perfect separator: its direction is the penalty’s tie-break among separating hyperplanes, and it drifts away from the mean-difference direction as the sample grows (cosine 0.97 to 0.63). With the reproducible direction, three pre-registered ablations test the linear account of the channel. Removing the emotion direction from the sender’s span at every readable layer leaves 91% of the transfer; a rank-5 affect subspace fails its own instrument check; and a fitted, label-free control blocks transfer where the field’s norm-matched random direction is inert, while a fitted label-free rank-1 direction leaves the emotion direction’s block of afraid (25%) and sad (29%) intact. Transfer is real, the linear rank-1 channel carries at most a third of it for two emotions, and ablation claims about concept directions need controls fitted by the same estimator, not random vectors.

1 Introduction

Systems of several language-model agents built from one model family are now common, and the agents talk to each other. If one of them is in an emotional state, does anything of that state reach the others? A transfer that changes the receiver’s internal representation would be invisible in transcripts and visible only in activations, which makes it a question for representation-level tools: steering vectors to induce a state in the sender [4, 7–9], probes to read it in the receiver, and ablations to test what carries it.

Those tools rest on one object, a direction in activation space. This paper’s single idea is that the direction has to be trusted twice, once to read and once to intervene, and that both uses fail in ways the usual checks miss. Read with a logistic-regression coefficient row, the state looks like a moving target: the direction does not reproduce across independent fits, and we show the reason is not the sample size or the penalty but separability, so the fitted row is a tie-break rather than an estimate. Intervene with the reproducible direction and the standard control, a norm-matched random vector, stays inert while a control fitted by the same estimator with the labels destroyed can block transfer as strongly as the emotion direction does.

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The evidence is three pre-registered experiments on the channel, each critiqued blind before its wording entered this paper, on top of an estimator study across two models and three depths. The transfer itself is robust: five of six emotions on each model, in three runs that reproduce each other bit for bit on the 27B. The channel is not what the linear picture suggests. Removing the emotion direction everywhere it can be read leaves a median 91% of the transfer, and only afraid and sad lose a fifth to a third; a rank-5 affect subspace cannot even be used as an instrument, because it shifts the readout used to check it; and a fitted label-free direction of rank 1 is inert while a fitted label-free frame of rank 5 is not.

The practical consequence is a change to the controls register of any ablation study of a concept direction: beside the random direction, run the same estimator on permuted labels, match rank and footprint if the intervention is more than rank 1, and check the manipulation with an instrument that does not share the ablation’s target. The scientific consequence for affect transfer is narrower than we hoped: the transfer exists, the rank-1 direction carries part of it for two emotions, and what carries the rest is open.

Contributions.

- Dose-dependent affect transfer into the receiver’s representation for five of six emotions on two models, with common-random-number sampling and scenario-blocked inference (Section 4).
- Disjoint-half reproducibility of probe directions as the missing quantity: logistic 0.57 / 0.29 against difference-of-means 0.97 / 0.94, unchanged by a fixed per-sample penalty, with the mechanism (perfect separation at every n ; the direction is the penalty’s tie-break and drifts from the mean difference) measured directly (Section 3).
- Three pre-registered ablations of the channel with their verdicts as registered: rank-1 insufficiency (H1), rank-5 instrument failure, and specificity of the rank-1 block against a fitted label-free control (H3), plus a replication of two designs on the second model (Section 5).
- A demonstration that fitted label-free controls bite where random ones do not, and the resulting rule for controls (Section 6).

Every number in this paper is registered with a pointer to the committed script and result file that produced it; the identifier appears in monospace after the number. Seven of our own interpretive sentences were withdrawn under blind review and are absent here.

Relation to prior work. That logistic probe directions can be unstable is not our discovery. Marks and Tegmark [5] identified the deficiency and proposed mass-mean probing; Gao et al. [2] name directional stability as a requirement for probe-then-steer pipelines and propose ridge-tuned logistic regression, reporting robustness of 0.88–0.99 under 20% training-data ablations. Our contribution is the quantity: reproducibility across *disjoint* halves, on the same estimator family, is 0.41–0.57 on the 27B (`stabdiag.logreg_27b_600`, `a2.focus_logreg_split_half`) and 0.29 on the 8B (`a2_8b.*`), and it does not rise with the sample size while difference-of-means rises to 0.94–0.98 on the same splits. Two overlapping 80% subsamples share most of their examples; two disjoint halves share none, and the gap between the two numbers is the finding. Braun et al. [1] corroborate from the other side, noting that steering vectors from few samples vary widely, without fitting twice. Herbst et al. [3] report mean-difference and logistic directions as near-identical (0.98–0.99) for compassion; in a quantity closer to theirs than our split-half battery, though not identical to it, we find 0.43 and 0.33–0.37. Raju [6] measure stability of a representation across feature dimensions, a different axis.

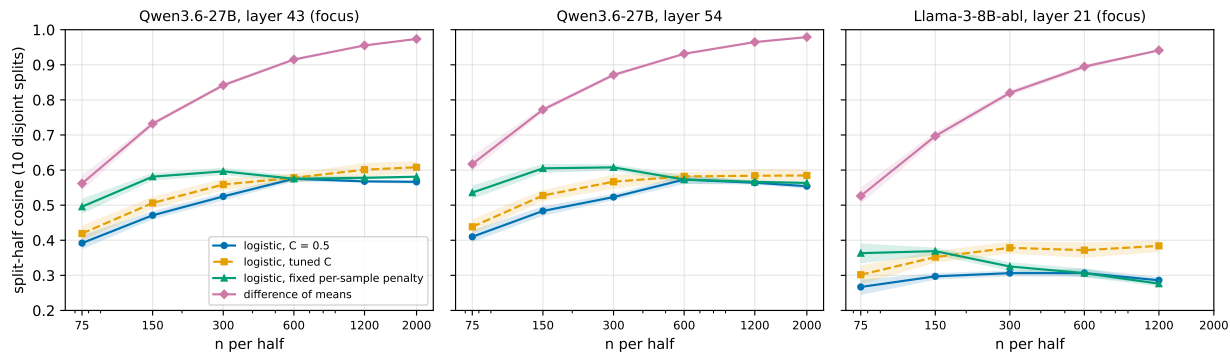


Figure 1: **Logistic directions do not reproduce; difference-of-means does.** Split-half cosine of the fitted direction over ten disjoint splits against n per half, at two depths of the 27B and the focus layer of the 8B. Difference-of-means rises toward 1 with n ; the three logistic variants plateau near 0.6 (27B) or fall to 0.28 (8B), including the one whose per-sample penalty is held fixed. Bands are bootstrap CIs over splits.

2 Setup

Two agents, A and B, converse in 29 fixed scenarios. A’s message is generated under a steering vector added at layer 42 of Qwen3.6-27B (revision 6a9e13bd) at doses 0, 0.33, 0.67 and 1.0 of a fixed scale, with three replicate seeds; B’s reply is then sampled with common random numbers across experimental arms, so that sampling noise shared between two arms cancels inside each paired contrast. Two readouts: B’s *present-emotion projection*, the projection of B’s reply activations on a probe fitted on one half of a 1,615-item probe pool while steering and ablation directions come from the disjoint other half; and a generation-side forced-choice readout of A’s span, used as the manipulation check. Every dose-response slope is an ordinary least-squares slope over the four doses on per-scenario means; every confidence interval is a scenario-blocked bootstrap (29 clusters, 5,000 draws), and every between-arm contrast is scenario-blocked and dose-paired. The second model is Llama-3-8B-Instruct-abliterated (revision dd67dd05), used for the estimator results and for replications of two ablation designs. All code runs from one vendored core; every number below was produced by a committed script from a committed result file.

3 The estimator: reproducibility and its mechanism

Disjoint-half reproducibility. We fit the same estimator on two disjoint halves of n items each, ten independent splits, and report the mean per-class cosine between the two fitted directions (Figure 1). At the 27B focus layer with $n = 2000$ per half, difference-of-means reaches 0.974 and logistic regression ($C = 0.5$) 0.566 (a2.focus_dom_split_half, a2.focus_logreg_split_half); at layer 54, 0.979 against 0.554 (a2f54.*); on the 8B focus layer at $n = 1200$, 0.941 against 0.286 (a2_8b.*). Tuning C per fit by cross-validation reaches 0.61 on the 27B (a2f.focus_tuned_c_does_not_close_gap) and 0.37–0.41 on the 8B.

Holding the per-sample penalty fixed does not help. A natural objection is that a fixed C changes the effective penalty as n grows. With $C_n = 0.5 N_{\text{ref}}/n$, so that the per-sample penalty is constant, the logistic curve shows no net rise from $n = 150$ to 2000 at two anchors on the 27B focus layer (0.581 at both ends for the 600 anchor, with a significant dip at 600 and a partial recovery;

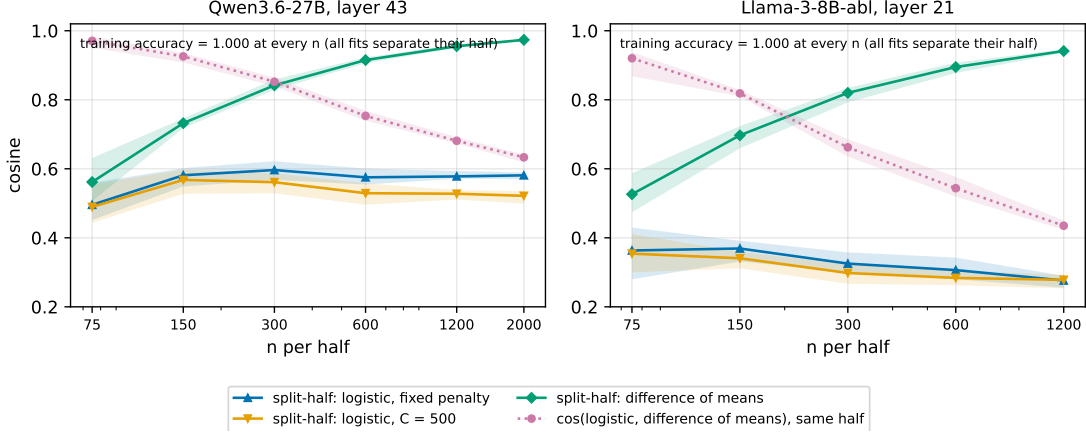


Figure 2: **The logistic direction is a separator’s tie-break that drifts away from the mean difference.** Within one half, the cosine between the fixed-penalty logistic direction and the difference-of-means direction (dotted) falls as n grows, while difference-of-means becomes reproducible across halves and the logistic direction does not, under the fixed schedule or a $1000\times$ weaker penalty alike. Every fit has training accuracy 1.000.

a2f.focus_logreg_fixed_lambda_flat, a2f2.lam_anchor2000_no_rise), one drop at $300 \rightarrow 600$ and then flat at layer 54 (a2f54.lam_declines_anchor600), a rise at layer 16, and on the 8B a peak at 150 followed by a fall to 0.276 at 1200 (a2f8.lam_declines). The picture is depth-specific, with no convergence over the reachable range ($n \leq 2000$, $d = 5120$): a proportional-regime statement, not an asymptotic one.

Why. Every multinomial logistic fit on every half at every n used, under the fixed schedule and $C = 0.5$, 50 and 500 alike (the two weak arms being one tolerance-limited solution), separates its training half perfectly: accuracy exactly 1.000 in every cell, arm and seed; worst single-fit log-loss 0.0055; at most 80 iterations (a2m.separable_every_n, a2m.8b_same_regime, a2m.v2_ten_seeds). The likelihood is saturated from $n = 75$: the direction is the penalty’s tie-break among separating hyperplanes. Weakening the penalty a hundredfold leaves the direction unchanged on the same half (cosine ≥ 0.96) and no more reproducible (0.52 against 0.58 at $n = 2000$; a2m.weak_penalty_same_direction). The separator starts near the mean-difference direction and departs from it as n grows, cosine $0.97 \rightarrow 0.63$ on the 27B (a2m.logistic_departs_from_dom) and $0.91 \rightarrow 0.43$ on the 8B (a2m.8b_same_regime), while difference-of-means becomes reproducible (Figure 2). The wording adopted from blind review: on separable activations the multinomial direction is the penalty’s tie-break among separating hyperplanes, not a likelihood estimate; its split-half reproducibility plateaus at 0.55–0.60 on the 27B and falls to 0.28 on the 8B across a 27-fold increase in n , while difference-of-means, an average rather than an extremum, rises to 0.94–0.98 on the same splits. Not claimed: that the fit sits at the max-margin limit (the weak arms are tolerance-limited), or a support-vector mechanism for the drift.

Against the near-identity claim. In a quantity closer to that of Herbster et al. [3] than our split-half battery, though still not identical to it (mean-difference against logistic, raw space, ten subsamples), the cosine is 0.43 on the 27B and 0.33–0.37 on the 8B, not 0.98 (a2f.caa_raw_vs_logreg_raw, a2f8.cross_estimator).

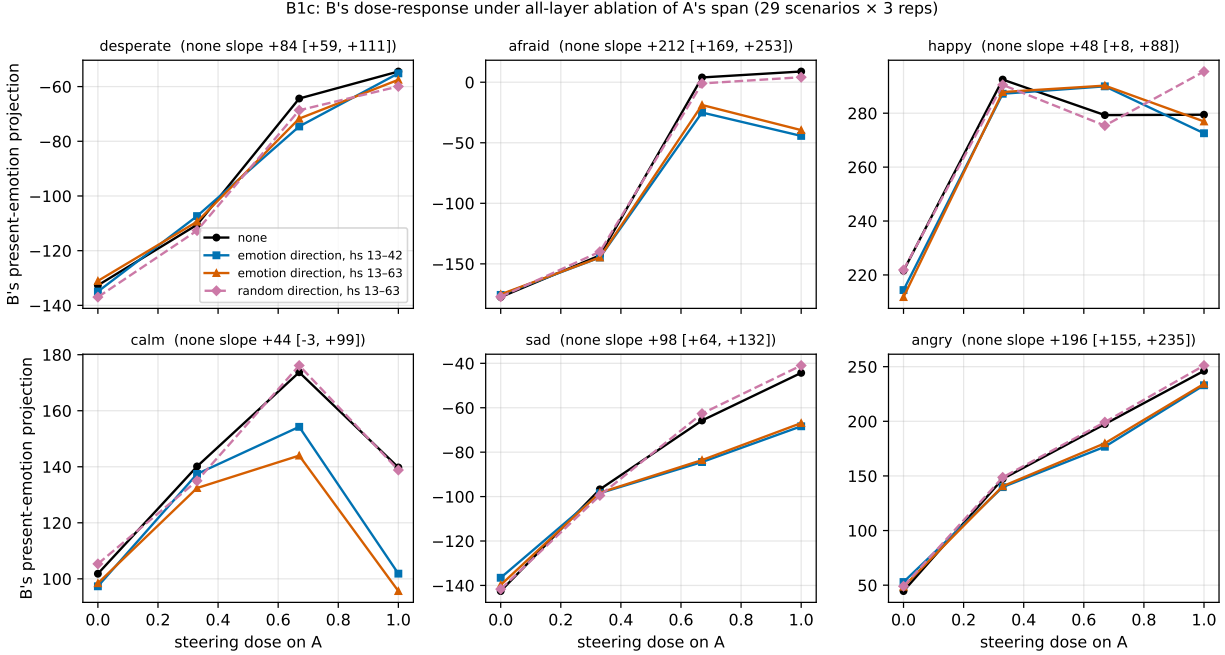


Figure 3: **Transfer is dose-dependent, and removing the emotion direction everywhere changes it little.** B’s present-emotion projection against A’s steering dose, per emotion, under no ablation and under the three ablation arms of B1c (29 scenarios $\times$ 3 replicates). Titles give the none-arm slope with its scenario-blocked CI; the ablation arms separate from the none arm only for afraid and sad.

4 Transfer

B’s present-emotion projection rises with A’s dose for desperate, afraid, happy, sad and angry, with none-arm slopes from +48 to +212 and CIs excluding zero; calm’s (+44 [−3, +99]) never does (b1c.verdict_H1; Figure 3). The same slopes appear, bit for bit, in the three runs that share the pool and directions (b1c.partial_block_replicates, b1d.replicates_b1c_arms), and the manipulation check on A passes for all six emotions in each. Happy’s is the weakest of the five (+48 [+8, +88]). On the 8B the same five emotions transfer (b1c8.transfer_replicates), with a steering manipulation half as strong and less specific, so only the pattern, not the size, is compared across models.

5 Three pre-registered ablations

B1c: the rank-1 direction (verdict H1). Projecting the difference-of-means emotion direction out of A’s span at every hidden state from 13 to 63 during B’s prefill removes a median 9% of the dose-response, CI [4%, 25%] over five testable emotions (b1c.verdict_H1, b1c.median_fraction). Afraid (21%) and sad (33%) are the only significant blocks (b1c.partial_block_replicates). Extending the ablation to layers 43–63 changes no dose-response detectably, with the extra blocking bounded under 0.08 for four emotions, a driver-declared contrast (b1c.no_gain_beyond_layer42). A unit-norm random direction, removing twice the norm, leaves five of six slopes within 5% of the none arm and moves happy by +28% with overlapping CIs, descriptively, since no random-versus-none contrast was computed (b1c.random_control_inert). The blocked fraction tracks how much readable affect the

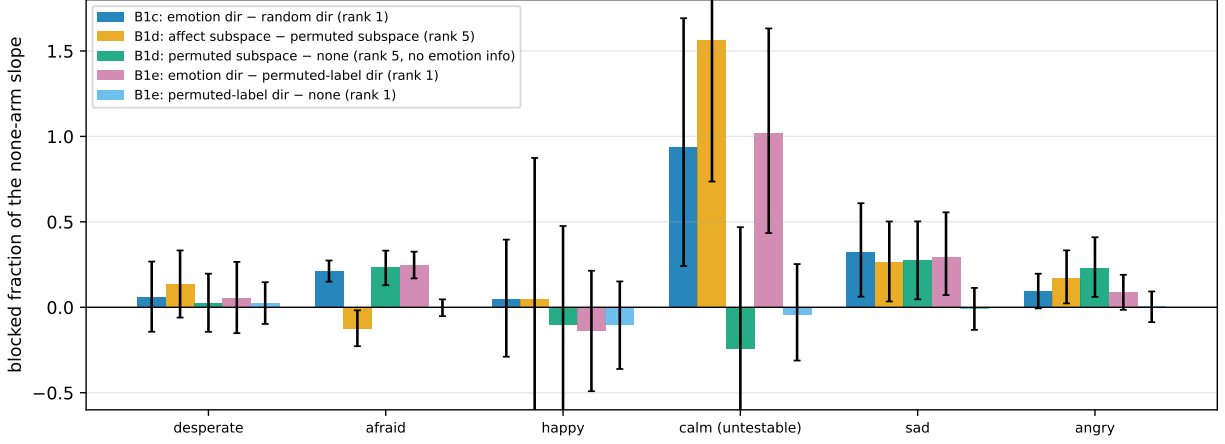


Figure 4: **Which ablations block, and which controls bite.** Blocked fraction of the none-arm slope, per emotion, for the decision contrast of each ablation experiment and for the two label-free controls against baseline. The fitted rank-5 permuted subspace (third bar) blocks afraid, sad and angry; the fitted rank-1 permuted direction (fifth bar) blocks nothing. Error bars are the CI-implied ranges. Calm is untestable (its none-arm slope includes zero).

ablation removes ($r \approx 0.85$ over five points; blocked-per-removed ≈ 0.27 ; `b1c.readout_residual`): the rank-1 direction is insufficient to carry the transfer.

B1d: the rank-5 subspace (verdict instrument_failed). Projecting out the rank-5 class-mean affect subspace, which captures 16–21% of the pool’s variance and has cosine 0.71–0.81 to the top-5 principal components (`b1d.frame_geometry`), removes at least 80% of the readable affect for only three of six emotions and shifts the readout at dose 0 (desperate 0.11→0.50; `b1d.readout_baseline_shift`): the instrument and the ablation share a target. A permuted-label subspace built by the same estimator, with no emotion information and a third of its span in common with the affect frame, reduces the transfer for afraid (−49.5), sad (−26.7) and angry (−44.3), as much as the emotion direction did (`b1d.perm_control_blocks`; unadjusted CIs), while a random 5-frame is inert. The emotion-specific increment over that control is 0.14 [−0.03, 0.29], and no per-emotion contrast clears the registered BH-FDR rule ($q = 0.051$; `b1d.median_fraction_descriptive`); rank 5 differs from rank 1 in opposite directions, less blocking for afraid (+29.3) and more for angry (−60.6) (`b1d.rank5_vs_rank1`).

B1e: a fitted, label-free rank-1 control (verdict H3). The same estimator on permuted labels gives a rank-1 direction with cosine 0.004 to the emotion direction (measured before launch, not stamped in the file) and $0.85\times$ its footprint; removed at all 51 layers it changes no emotion’s dose-response and removes no readable affect (`b1e.permdir_inert`). Against it the emotion direction blocks afraid by 25% [17%, 33%] and sad by 29% [7%, 56%] (BH-FDR q 0.001 and 0.028; `b1e.verdict_H3`). The top principal direction of A’s span, with $5.8\times$ the footprint, blocks angry by 61% (and calm, knife-edge on an untestable emotion) while leaving afraid and sad untouched even though it removes 79% and 71% of their readable affect (`b1e.pc1_dissociation`): readable-affect removal and blocking dissociate. B1d’s rank-5 permuted frame blocks where B1e’s rank-1 permuted direction does not; they differ in rank and in footprint (6.5 against 0.5 per position), and which matters is untested. The control’s footprint is below the emotion direction’s, so the footprint caveat stands; the secondary readout on B’s reply agrees for sad and not for afraid (Appendix A).

Table 1: The three ablation experiments on the 27B, decision contrast per emotion (blocked fraction of the none-arm slope; significant contrasts in bold). Calm is untestable.

	desperate	afraid	happy	calm	sad	angry
B1c: emotion dir – random dir	0.06	0.21	0.05	(0.94)	0.33	0.09
B1d: affect subspace – permuted subspace	0.14	−0.13	0.05	(1.56)	0.26	0.17
B1d: permuted subspace – none	0.03	0.23	−0.10	(−0.24)	0.27	0.23
B1e: emotion dir – permuted-label dir	0.06	0.25	−0.14	(1.02)	0.29	0.09
B1e: permuted-label dir – none	0.02	−0.00	−0.10	(−0.04)	−0.01	0.00

Replication on the 8B. B1c’s design blocks the same two emotions to significance on the 8B (15%, 31%; `b1c8.verdict_and_block`), with the other three too wide to tell agreement from disagreement; its manipulation-check rule returned `instrument_failed` because the upper ablation window changes the readout by nothing measurable there and the strict criterion has no tolerance band. B1e’s design returns H3 on the 8B with sad blocked by 36% (q 0.002, secondary readout agreeing) and afraid by 14% (q 0.08, no block against baseline); the control is matched only on average (sad 0.86 $\times$), both directions remove about the isotropic norm, and B1c’s random direction on that model reproduced the contrasts, so the run adds no discriminating power there. Only sad replicates across models on both outcomes; afraid is unreplicated (`b1e8.verdict_H3_matched`).

6 What the controls teach

Norm-matched random and orthogonal directions, the field’s standard controls, were inert in every run; so was a fitted, label-free rank-1 direction. What blocked without emotion information was a fitted rank-5 frame with ten times the emotion direction’s footprint. Any ablation claim about a concept direction therefore needs, beside the random control, a control fitted by the same estimator on permuted labels, and if the intervention is more than rank 1, a control matched in rank and footprint; and the manipulation check must not share its target with the ablation, or the check fails in the way B1d’s did.

7 Limitations and what is not claimed

One probe pool and one direction estimate underlie all ablation runs on the 27B; the A-span readout is forced-choice and shares a target with the subspace ablation; calm is untestable at this dose grid; B1d’s per-emotion contrasts do not clear BH-FDR; the median CIs are scenario-only bootstraps and do not include emotion-to-emotion spread. Provenance limits and the defects recorded under review are listed in Appendix A. Not tested: the token channel (three rewrite designs failed their own manipulation check: the neutral rewrite kept 67–98% of the readable affect and the affect-preserving rewrite changed the transfer itself; `b1c.text_arms_uninterpretable`), the behavioural reach, layers below 13, and $n > d$.

8 Conclusion

The idea to take away is not that affect transfers between agents, though it does, nor that logistic probes are unstable, which was known. It is that the two failures we found are the same failure seen twice. A logistic direction on separable activations is decided by the penalty’s tie-break, so

two independent fits agree only as far as their tie-breaks do; and a control direction is informative only if it is decided the same way the intervention direction was. Random vectors are decided by nothing, which is why they are inert and why they miss what a fitted control catches. Any study that reads a concept with a probe and intervenes with its direction should therefore report disjoint-half reproducibility of the direction, use an estimator that is an average rather than an extremum, and run its ablations against a control fitted by the same estimator on permuted labels. For affect transfer specifically, the linear rank-1 direction carries a fifth to a third of the effect for two emotions and none of it for the others, and the next experiment is one that separates the receiver re-reading the sender’s tokens from the receiver reading the sender’s residual stream.

Reproducibility statement. Code, pre-registrations, result files and the claim registry are in the accompanying repository (research-use licence). Models: Qwen3.6-27B revision 6a9e13bd and Llama-3-8B-Instruct-abliterated revision dd67dd05, run with torch 2.11.0+cu130 and transformers 5.14.1 on one NVIDIA RTX PRO 6000 (96 GB) per run; a 27B ablation experiment takes about two hours (432 or 288 arm-rows at 19–21s each), the 8B one under thirty minutes. Seeds: pool generation seed 0; per-generation seeds are a CRC32 of (seed, emotion, dose, replicate, arm, part) with the arm field empty for the receiver’s reply so that arms share random numbers; ten split seeds (0–9) for every split-half number; bootstrap seed 0 with 5,000 draws. Hyperparameters: doses $\{0, 0.33, 0.67, 1\}$ at a fixed activation-RMS scale, three replicates, 29 scenarios, 110 new tokens per message at temperature 0.9; logistic regression with lbfgs, max_iter 3000, $C \in \{0.5, C_n = 0.5 N_{\text{ref}}/n \text{ with } N_{\text{ref}} \in \{600, 2000\}, 50, 500\}$ and cross-validated C on a decade grid scored for accuracy; split-half $n \in \{75, 150, 300, 600, 1200, 2000\}$ per half. Statistics: scenario-blocked percentile bootstraps because scenarios are the sampling unit; between-arm contrasts paired within scenario and dose; Benjamini–Hochberg over the testable emotions where a per-emotion rate is reported; every interval in the paper is such a bootstrap and every count of runs is stated. The probe pools and scenario lists are committed with the results.

A Details and defects recorded under review

Each item below is in the registry note of the claim it qualifies.

- *B1c.* The contrast between the 30-layer and 51-layer windows was declared in the driver, not in the pre-registration; its null is unpowered for desperate and calm. Calm’s contrast is the largest and is dropped by the testability rule. Afraid’s top-dose readout is non-monotone. The pre-registration stamp is empty because the driver resolved its path relative to the wrong directory; the document was on the machine one second before the run started. The pool has 1,615 items against the registered 2,160; the false-discovery rate ran over five emotions rather than six.
- *B1d.* The permuted-label subspace removed only 6.5% of afraid’s readable affect and still blocked its transfer by 49.5, so overlap with the readable-affect direction does not explain its blocking; recomputing the instrument share with each arm’s own floor keeps three failures but changes which emotions fail. The median’s confidence interval resamples scenarios only. No per-emotion contrast clears the registered false-discovery threshold ($q = 0.051$).
- *B1e.* The control removes less norm than the emotion direction ($0.84\times$ afraid, $0.77\times$ sad) and less than B1c’s inert random direction (1.22), so it is a fitted label-free null, not a footprint-matched one; the $0.85\times$ – $5.8\times$ footprint band, where B1d’s frame blocked, is unsampled. The rule tests the point fraction and sad’s range starts at 0.07. The secondary readout on B’s reply

contradicts afraid’s block and agrees for sad. The control’s cosine to the emotion direction (0.004) was measured by the pre-launch verifier and is not stamped in the result file.

- *8B replications.* Both are out-of-pre-registration replications (the documents name the 27B and its layers). The steering manipulation is half as strong and less specific on the 8B, afraid’s readout sits below chance, and slopes are unnormalised probe scores, so only patterns are compared. B1c’s manipulation-check rule uses a strict $\geq$ on nested windows and fails by chance when the upper window changes nothing, which it does not on this model. In B1e on the 8B the control is matched on average only (sad $0.86\times$), both directions remove about the isotropic norm, and B1c’s random direction already reproduced the contrasts; afraid does not block against baseline there. Only sad’s block replicates across models on both outcomes.
- *Estimator study.* The weak-penalty arms ($C = 50$ and 500) give the same solution to three decimals, so they are tolerance-limited and bound nothing about the $C \rightarrow \infty$ limit; the max-margin identification and the support-vector mechanism are not claimed. Cross-estimator cosines use subsamples that overlap by about half the pool, so their intervals overstate precision. The layer-43 and layer-54 files come from three driver builds; the shared cells agree bit for bit.

References

- [1] Joschka Braun, Carsten Eickhoff, David Krueger, Seyed Ali Bahrainian, and Dmitrii Krasheninikov. Understanding (Un)Reliability of steering vectors in language models. *arXiv preprint arXiv:2505.22637*, 2025.
- [2] Ziqi Gao, Yaotian Zhu, Qingcheng Zeng, Xu Zhao, Ziqing Wang, Feng Ruan, and Kaize Ding. RAPTOR: Ridge-adaptive logistic probes. *arXiv preprint arXiv:2602.00158*, 2026.
- [3] Niklas Herbster, Martin Zborowski, Alberto Tosato, Gauthier Gidel, and Tommaso Tosato. Activation steering for aligned open-ended generation without sacrificing coherence. *arXiv preprint arXiv:2604.08169*, 2026.
- [4] Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. *arXiv preprint arXiv:2306.03341*, 2023.
- [5] Samuel Marks and Max Tegmark. The geometry of truth: Emergent linear structure in large language model representations of true/false datasets. *arXiv preprint arXiv:2310.06824*, 2023.
- [6] Prashant C. Raju. Geometric stability: The missing axis of representations. *arXiv preprint arXiv:2601.09173*, 2026.
- [7] Nina Rimsky, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner. Steering Llama 2 via contrastive activation addition. *arXiv preprint arXiv:2312.06681*, 2023.
- [8] Alexander Matt Turner, Lisa Thiergart, David Udell, Gavin Leech, Ulisse Mini, and Monte MacDiarmid. Activation addition: Steering language models without optimization. *arXiv preprint arXiv:2308.10248*, 2023.

- [9] Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.